Introduction

The **Gym Management System** is a database-driven desktop application designed to efficiently manage gym operations such as member registration, trainer assignment, workout planning, body metrics tracking, and payment management.

The system replaces manual record keeping with a secure and structured database solution, improving data accuracy, accessibility, and operational efficiency.

Problem Statement

Traditional gym management relies on manual registers or spreadsheets, which leads to:

- Data redundancy and errors
- Difficulty in tracking member progress
- Poor payment monitoring
- Inefficient record maintenance
- Lack of analytics

A centralized digital database system is needed to streamline gym operations.

Objectives

- To develop an automated Gym Management System
- To efficiently manage member and trainer information
- To simplify membership and payment tracking
- To monitor fitness progress using body metrics
- To reduce manual work and improve accuracy
- To provide a user-friendly and interactive interface

Scope and Motivation

The scope of this project includes managing all core gym activities such as member records, trainer details, workout plans, payments, and equipment tracking. The motivation behind this system is the growing need for automation in fitness centers, where manual systems lead to inefficiencies, errors, and lack of proper tracking. This system aims to provide a centralized and efficient solution.

Methodology

The methodology of the Gym Management System follows a structured approach based on layered architecture and the MVC design pattern. The system is developed using Python, where PyQt5 is used for building the graphical user interface, and SQLite is used for database management. The development process includes designing the user interface, implementing business logic for handling operations such as member and trainer management, and integrating the database for data storage and retrieval. CRUD operations are used to manage all records efficiently. The system ensures proper data validation, error handling, and smooth interaction between different modules. This approach results in a well-organized, scalable, and maintainable application.

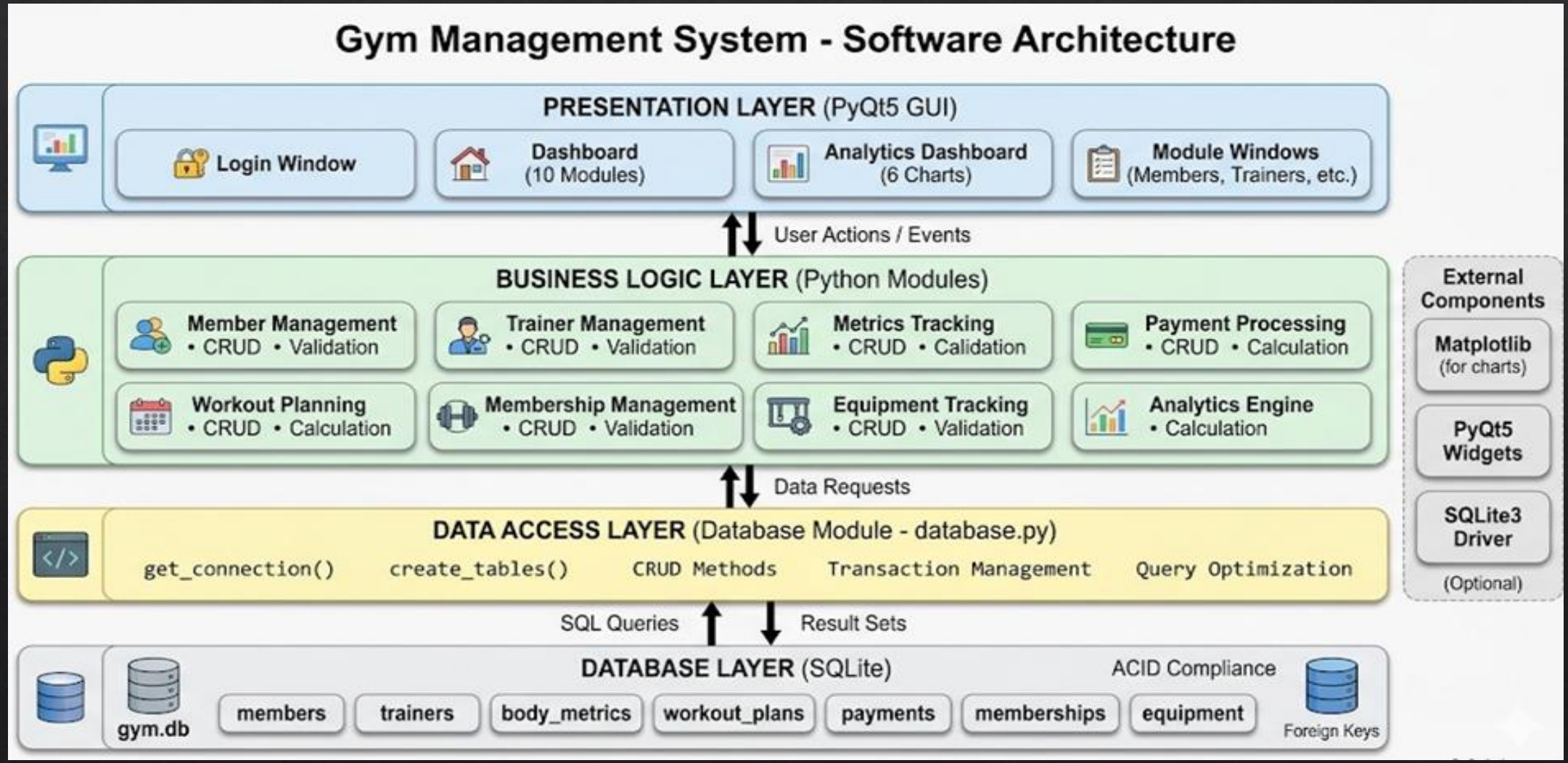
Proposed Solution

We propose a **Gym Management System** built using Python and SQLite that:

- Stores structured member and trainer data
- Tracks fitness metrics and BMI
- Manages memberships and payments
- Generates visual progress reports
- Maintains equipment records

The application ensures organized, reliable, and real-time data and analytics management.

Architectural Diagram

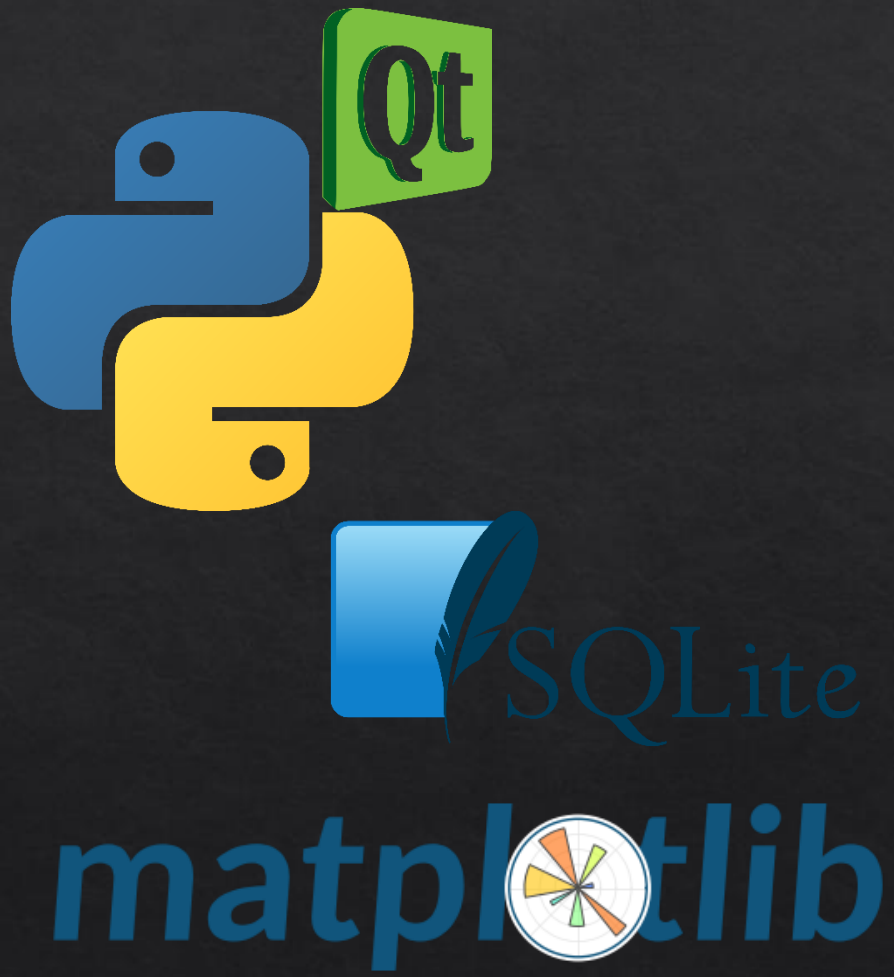


List of Modules

- **Login Module** – Handles authentication
- **Dashboard Module** – Provides navigation
- **Members Module** – Manages member details
- **Trainers Module** – Manages trainer information
- **Body Metrics Module** – Tracks fitness data and BMI
- **Workout Plans Module** – Assigns fitness programs
- **Payments Module** – Records transactions
- **Membership Module** – Manages plans
- **Equipment Module** – Tracks inventory
- **Analytics Module** – Provides real-time insights and charts

Implementation

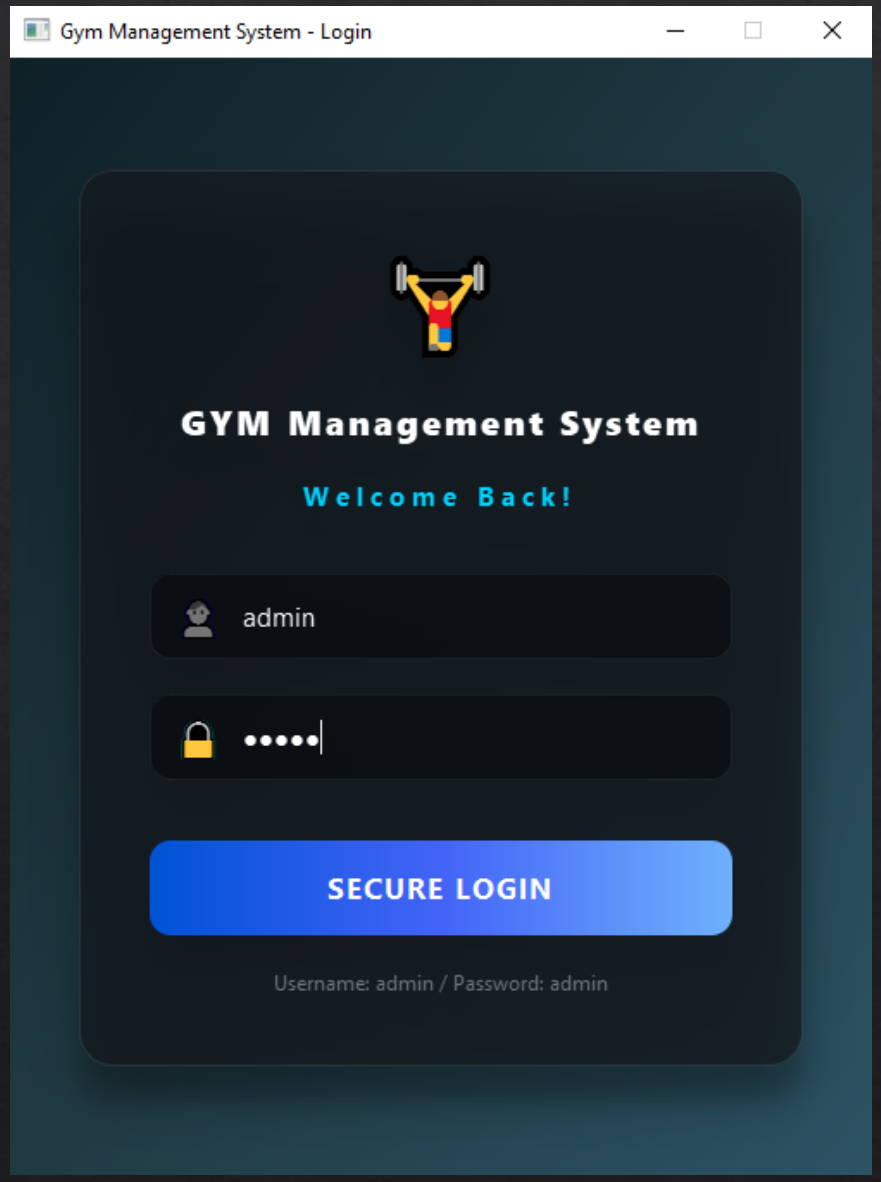
- ✓ The system is implemented using a desktop-based architecture
- ✓ **Front-End:** Developed using Python with PyQt5 for GUI design
- ✓ **Back-End:** Implemented using SQLite database for data storage
- ✓ CRUD operations are used to manage members, trainers, payments, and equipment
- ✓ Database connectivity is handled using Python's sqlite3 module
- ✓ Business logic is implemented using Object-Oriented Programming concepts
- ✓ MVC (Model-View-Controller) pattern is followed for structured development
- ✓ Matplotlib is used for generating charts and analytics



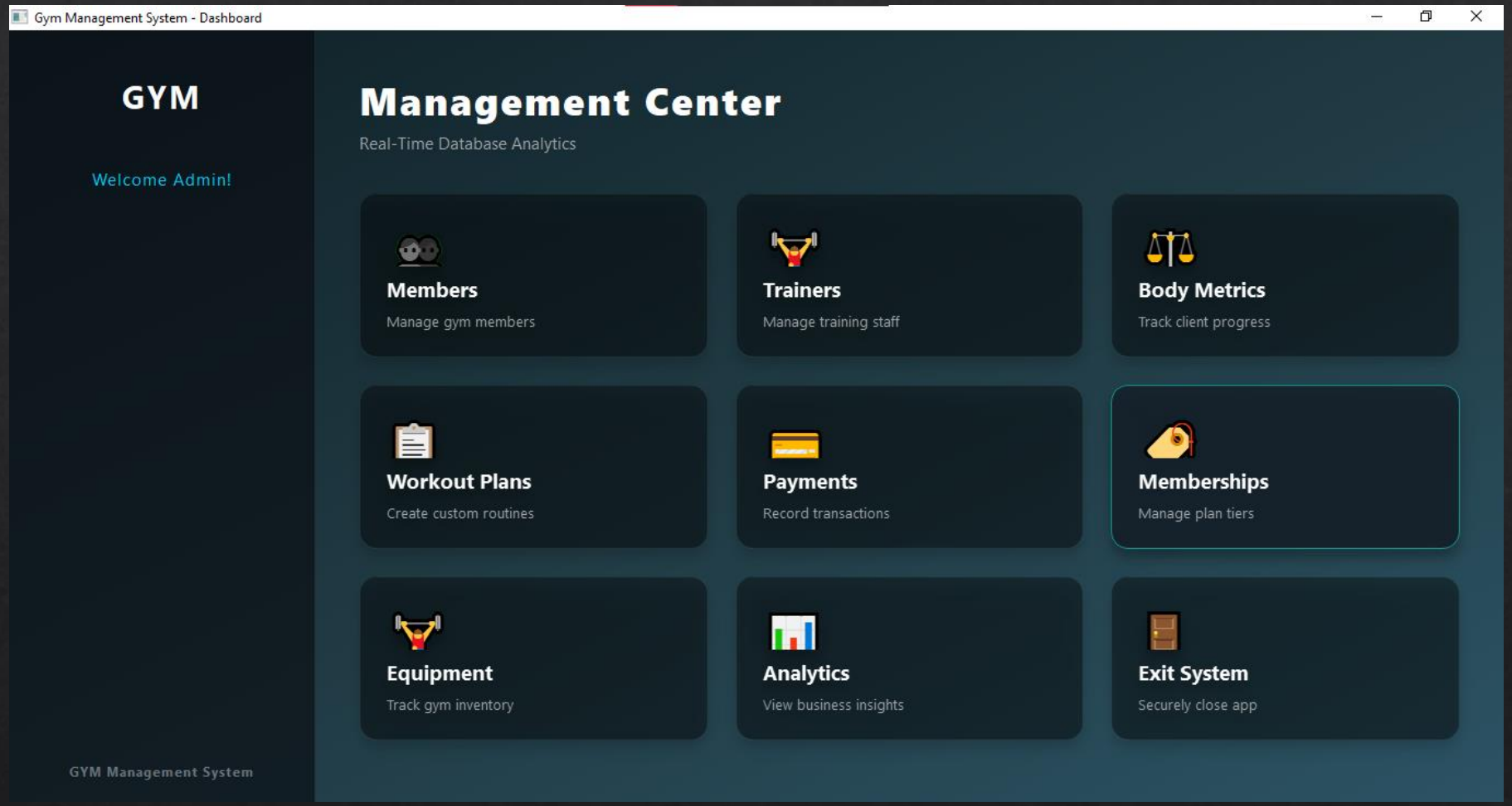
Sample Screenshots



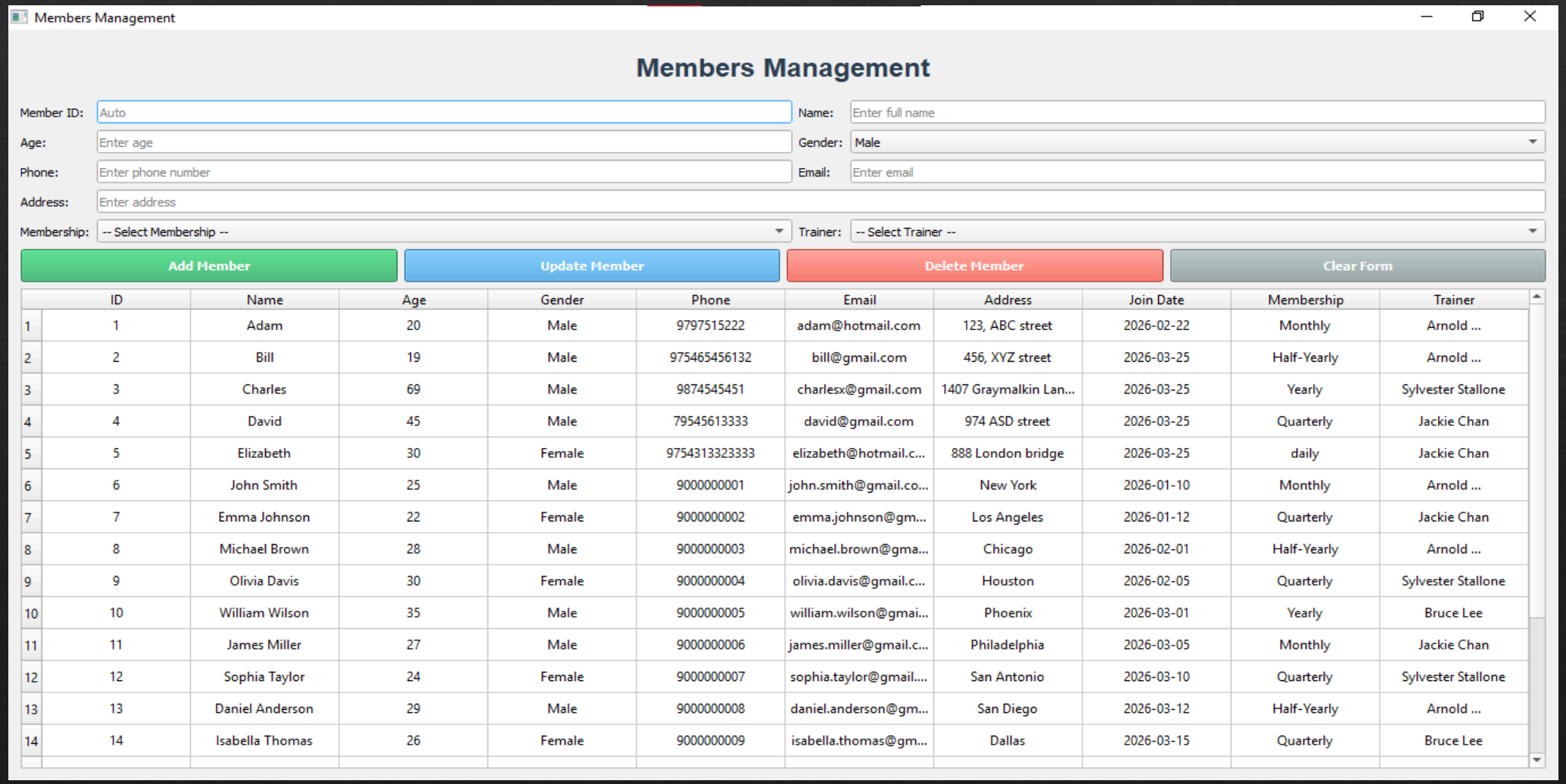
Login


Gym Management System - Login
GYM Management System
Welcome Back!
admin
SECURE LOGIN
Username: admin / Password: admin

Management Center

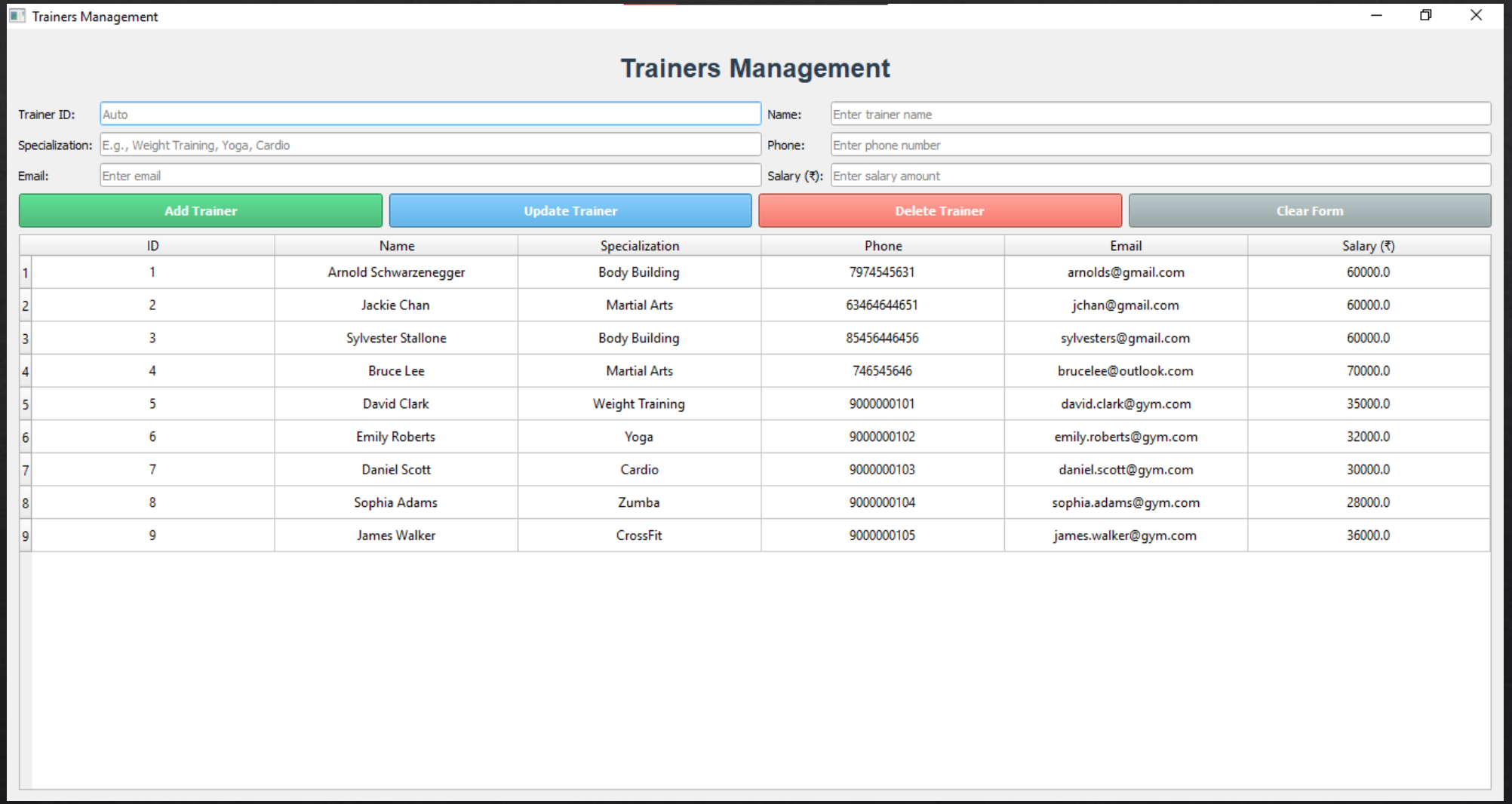


Members Management



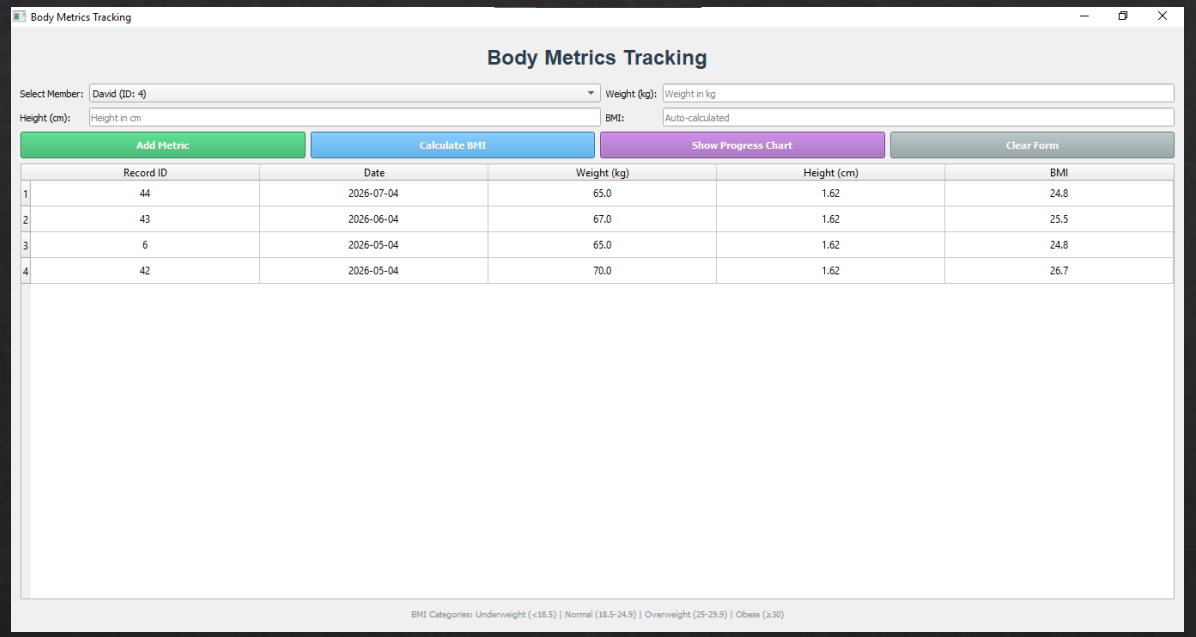
	ID	Name	Age	Gender	Phone	Email	Address	Join Date	Membership	Trainer
1	1	Adam	20	Male	9797515222	adam@hotmail.com	123, ABC street	2026-02-22	Monthly	Arnold ...
2	2	Bill	19	Male	975465456132	bill@gmail.com	456, XYZ street	2026-03-25	Half-Yearly	Arnold ...
3	3	Charles	69	Male	9874545451	charlesx@gmail.com	1407 Graymalkin Lan...	2026-03-25	Yearly	Sylvester Stallone
4	4	David	45	Male	79545613333	david@gmail.com	974 ASD street	2026-03-25	Quarterly	Jackie Chan
5	5	Elizabeth	30	Female	9754313323333	elizabeth@hotmail.c...	888 London bridge	2026-03-25	daily	Jackie Chan
6	6	John Smith	25	Male	9000000001	john.smith@gmail.co...	New York	2026-01-10	Monthly	Arnold ...
7	7	Emma Johnson	22	Female	9000000002	emma.johnson@gm...	Los Angeles	2026-01-12	Quarterly	Jackie Chan
8	8	Michael Brown	28	Male	9000000003	michael.brown@gma...	Chicago	2026-02-01	Half-Yearly	Arnold ...
9	9	Olivia Davis	30	Female	9000000004	olivia.davis@gmail.c...	Houston	2026-02-05	Quarterly	Sylvester Stallone
10	10	William Wilson	35	Male	9000000005	william.wilson@gmai...	Phoenix	2026-03-01	Yearly	Bruce Lee
11	11	James Miller	27	Male	9000000006	james.miller@gmail.c...	Philadelphia	2026-03-05	Monthly	Jackie Chan
12	12	Sophia Taylor	24	Female	9000000007	sophia.taylor@gmail....	San Antonio	2026-03-10	Quarterly	Sylvester Stallone
13	13	Daniel Anderson	29	Male	9000000008	daniel.anderson@gm...	San Diego	2026-03-12	Half-Yearly	Arnold ...
14	14	Isabella Thomas	26	Female	9000000009	isabella.thomas@gm...	Dallas	2026-03-15	Quarterly	Bruce Lee

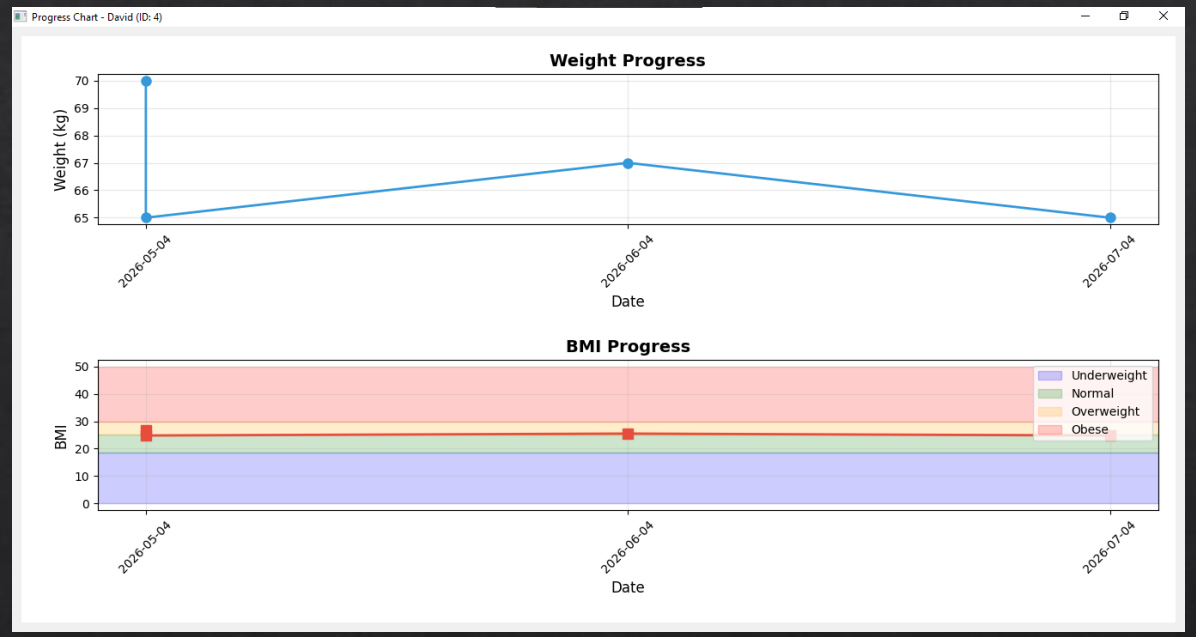
Trainers Management



	ID	Name	Specialization	Phone	Email	Salary (₹)
1	1	Arnold Schwarzenegger	Body Building	7974545631	arnolds@gmail.com	60000.0
2	2	Jackie Chan	Martial Arts	63464644651	jchan@gmail.com	60000.0
3	3	Sylvester Stallone	Body Building	85456446456	sylvesters@gmail.com	60000.0
4	4	Bruce Lee	Martial Arts	746545646	brucelee@outlook.com	70000.0
5	5	David Clark	Weight Training	9000000101	david.clark@gym.com	35000.0
6	6	Emily Roberts	Yoga	9000000102	emily.roberts@gym.com	32000.0
7	7	Daniel Scott	Cardio	9000000103	daniel.scott@gym.com	30000.0
8	8	Sophia Adams	Zumba	9000000104	sophia.adams@gym.com	28000.0
9	9	James Walker	CrossFit	9000000105	james.walker@gym.com	36000.0

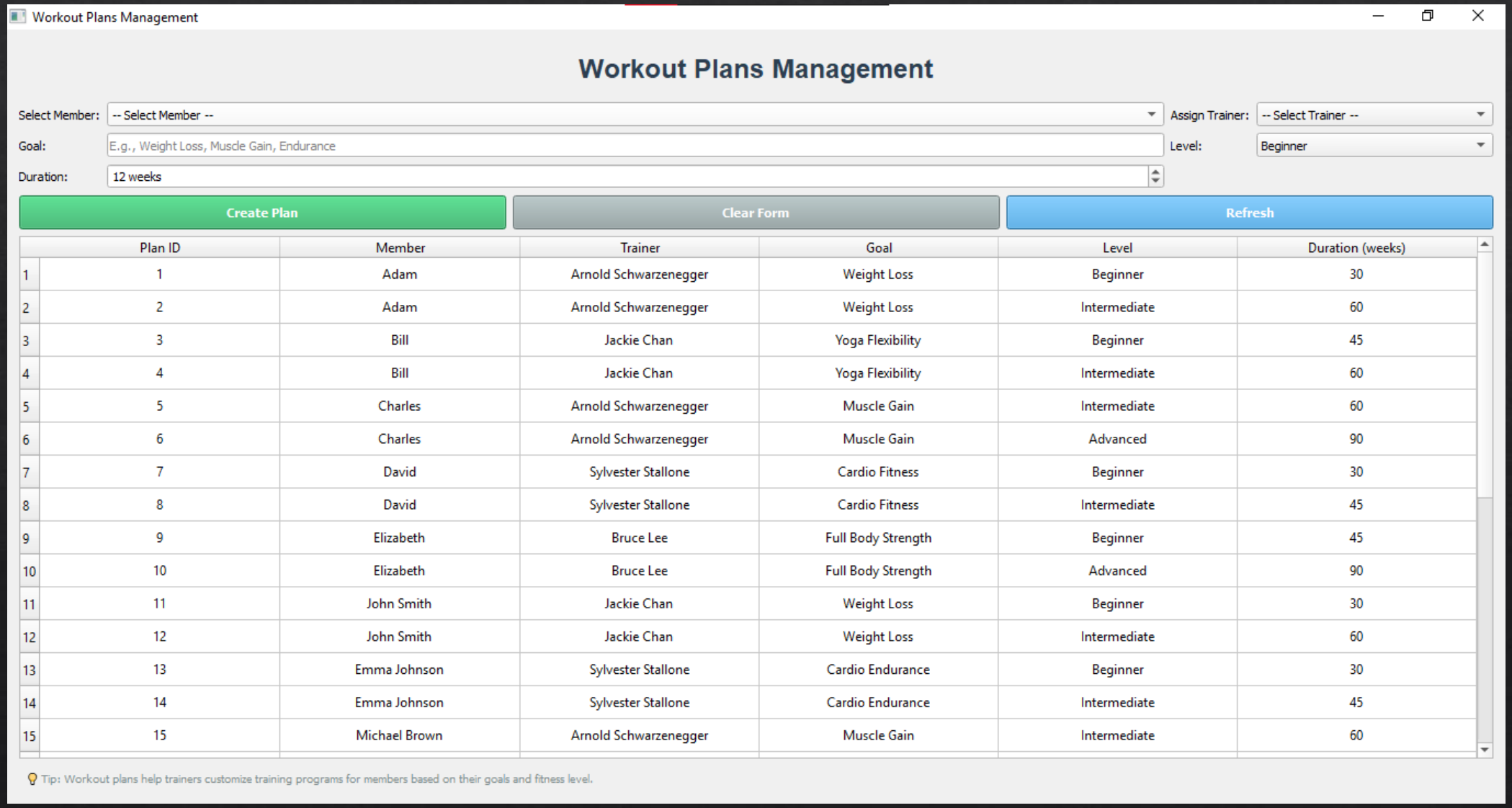
Body Metrics Tracking





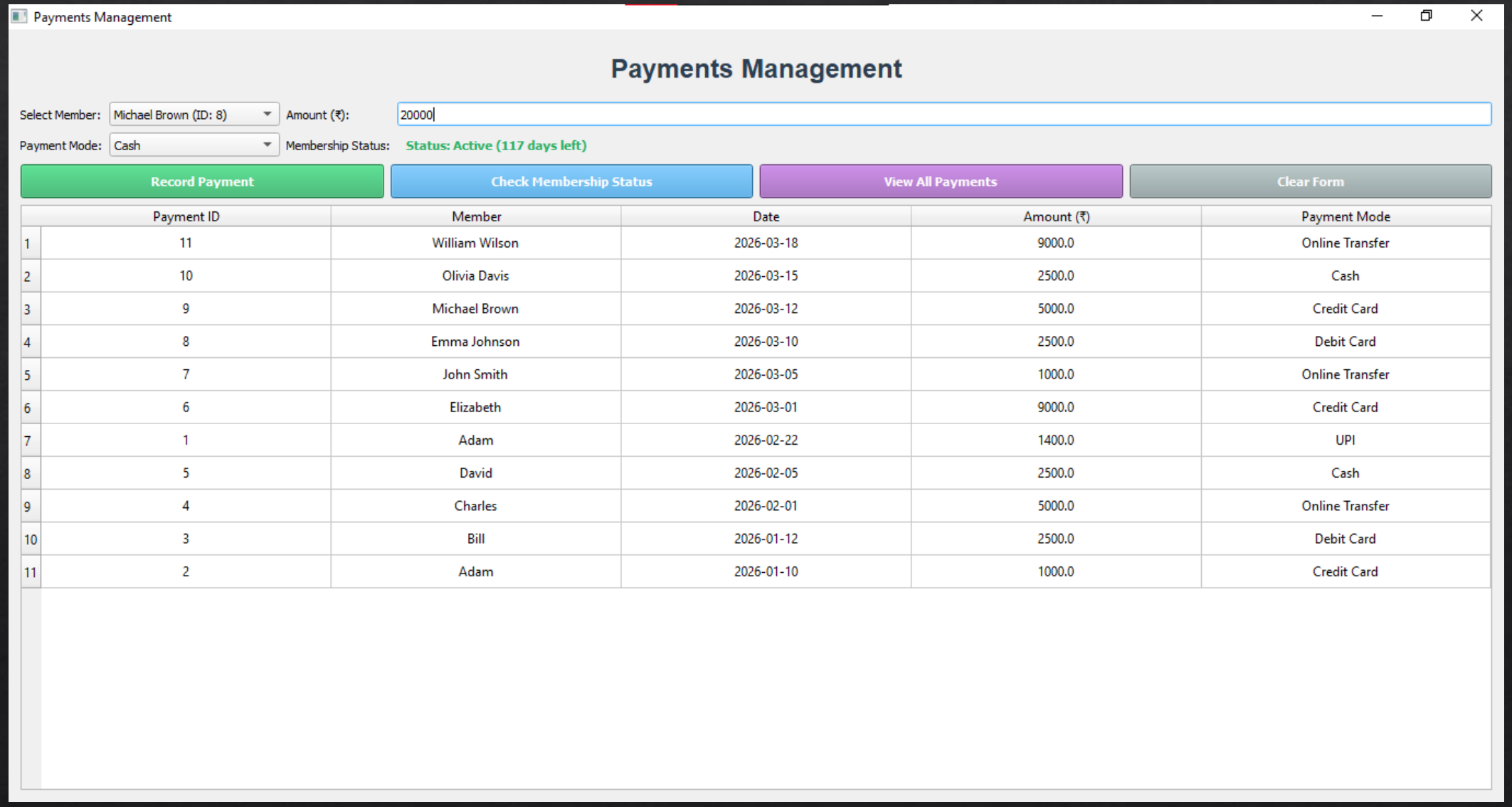
The Body Metrics Tracking module records and monitors the fitness data of members, including weight, height, and BMI. It maintains multiple entries for each member over time, allowing continuous tracking of their fitness progress. This helps trainers analyze improvements and adjust workout plans accordingly. The system ensures that all health-related data is stored in an organized and accurate manner, supporting effective performance evaluation and better fitness management.

Workout Plans Management



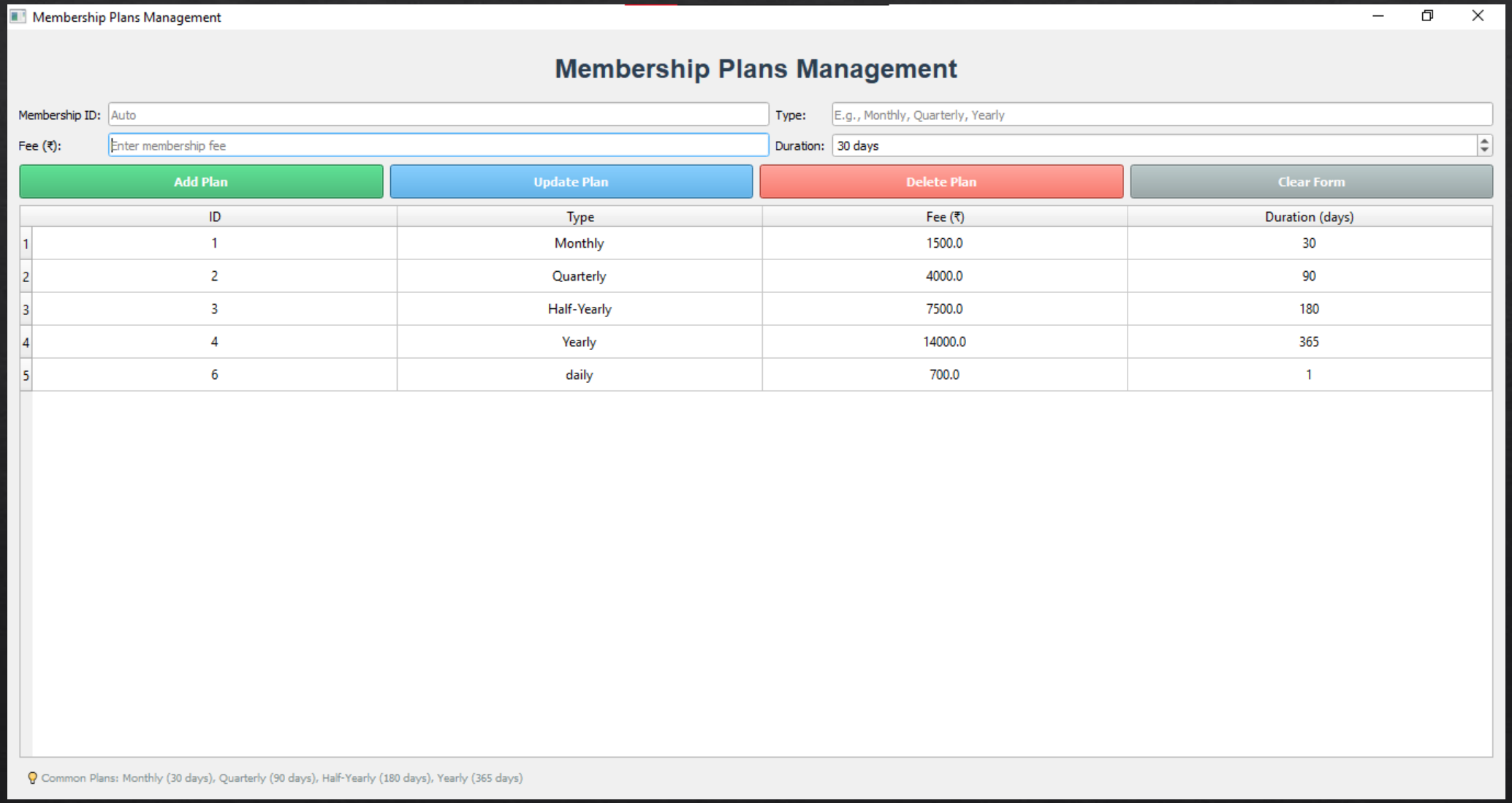
	Plan ID	Member	Trainer	Goal	Level	Duration (weeks)
1	1	Adam	Arnold Schwarzenegger	Weight Loss	Beginner	30
2	2	Adam	Arnold Schwarzenegger	Weight Loss	Intermediate	60
3	3	Bill	Jackie Chan	Yoga Flexibility	Beginner	45
4	4	Bill	Jackie Chan	Yoga Flexibility	Intermediate	60
5	5	Charles	Arnold Schwarzenegger	Muscle Gain	Intermediate	60
6	6	Charles	Arnold Schwarzenegger	Muscle Gain	Advanced	90
7	7	David	Sylvester Stallone	Cardio Fitness	Beginner	30
8	8	David	Sylvester Stallone	Cardio Fitness	Intermediate	45
9	9	Elizabeth	Bruce Lee	Full Body Strength	Beginner	45
10	10	Elizabeth	Bruce Lee	Full Body Strength	Advanced	90
11	11	John Smith	Jackie Chan	Weight Loss	Beginner	30
12	12	John Smith	Jackie Chan	Weight Loss	Intermediate	60
13	13	Emma Johnson	Sylvester Stallone	Cardio Endurance	Beginner	30
14	14	Emma Johnson	Sylvester Stallone	Cardio Endurance	Intermediate	45
15	15	Michael Brown	Arnold Schwarzenegger	Muscle Gain	Intermediate	60

Payments Management

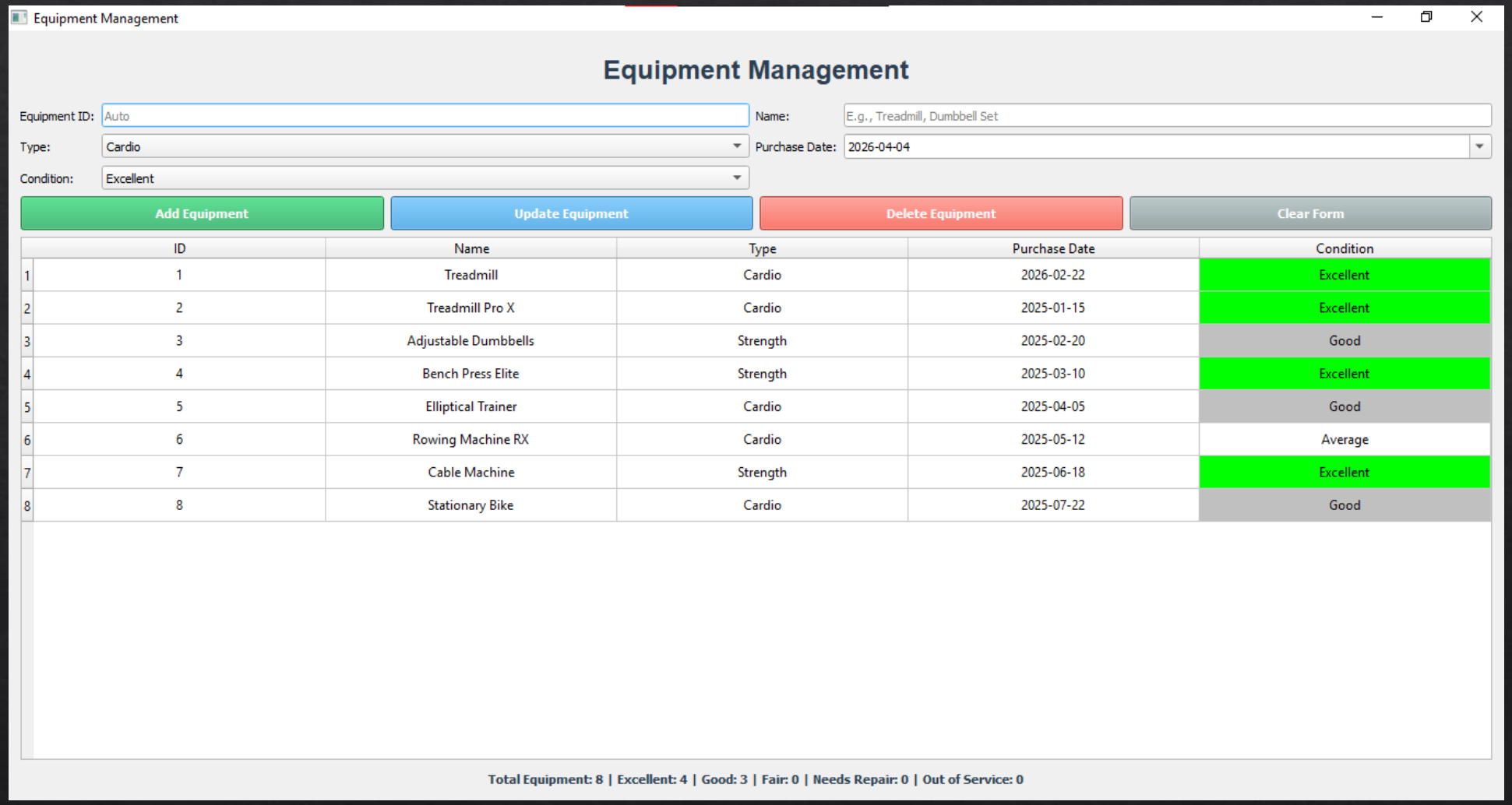


	Payment ID	Member	Date	Amount (₹)	Payment Mode
1	11	William Wilson	2026-03-18	9000.0	Online Transfer
2	10	Olivia Davis	2026-03-15	2500.0	Cash
3	9	Michael Brown	2026-03-12	5000.0	Credit Card
4	8	Emma Johnson	2026-03-10	2500.0	Debit Card
5	7	John Smith	2026-03-05	1000.0	Online Transfer
6	6	Elizabeth	2026-03-01	9000.0	Credit Card
7	1	Adam	2026-02-22	1400.0	UPI
8	5	David	2026-02-05	2500.0	Cash
9	4	Charles	2026-02-01	5000.0	Online Transfer
10	3	Bill	2026-01-12	2500.0	Debit Card
11	2	Adam	2026-01-10	1000.0	Credit Card

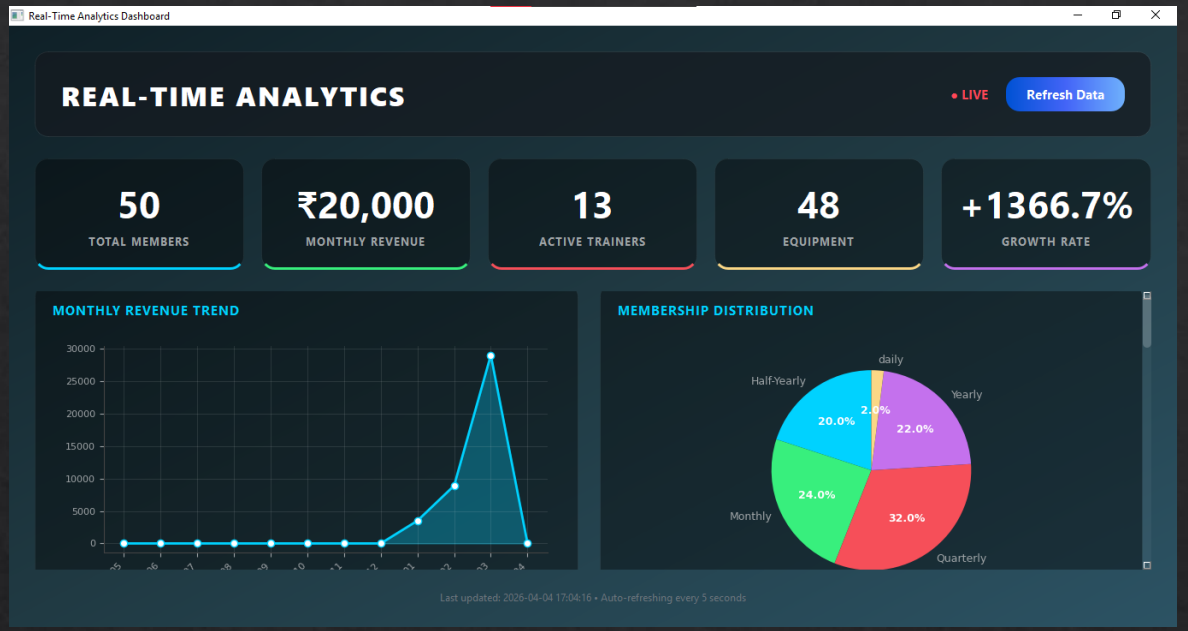
Membership Plans Management

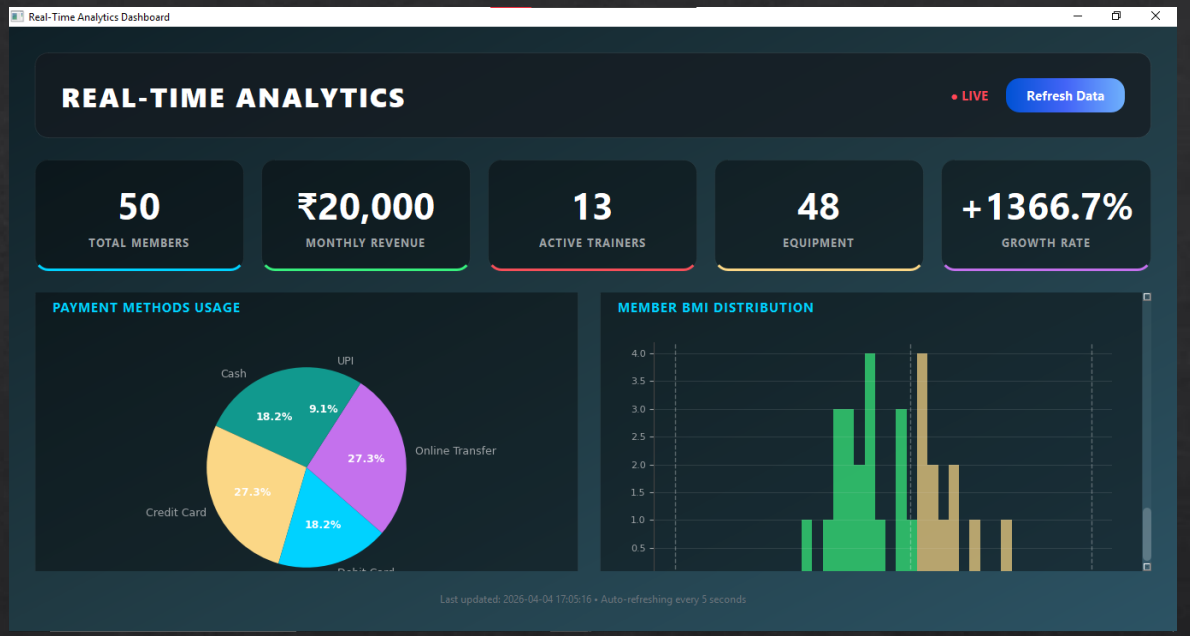


Equipment Management



Real-Time Analysis Dashboard





- Displays real-time insights of gym operations
- Visual representation of data using charts and graphs
- Tracks member growth and registration trends
- Shows revenue analysis based on payments
- Monitors fitness progress such as BMI and weight changes
- Helps in decision-making through data visualization

Results & Discussion

- User logs into the system
- Admin manages members, trainers, memberships, and equipment
- Member details are stored in the database
- Trainer is assigned to members
- Trainer creates and assigns workout plans
- System records body metrics (weight, BMI, etc.)
- Members make payments and transactions are stored
- Data is processed through business logic
- Reports and analytics are generated
- Results are displayed on dashboard and analytics screen

```
Login → Manage Data → Assign Trainer → Workout Plan → Store Data → Analytics → Dashboard
```

References

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- https://docs.python.org/3/library/sqlite3.html
- https://www.w3schools.com/sql/
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- https://pypi.org/project/PyQt5/
- https://matplotlib.org/stable/index.html
- https://pypi.org/project/matplotlib/

Conclusion

The Gym Management System was successfully designed and implemented as a desktop-based application. It effectively automates key gym operations such as managing members, trainers, payments, workout plans, and fitness tracking. The system improves efficiency, reduces manual effort, and ensures accurate data management. With its user-friendly interface and structured architecture, the application provides a reliable and scalable solution for modern gym management.